Gutenberg Atlas: Supplementary Materials

Corpus construction, composition, validation, and reuse demonstration

Release snapshot: 21 October 2026

Contents

1 Scope and reproducibility **1**

2 Corpus construction **2**

3 Release composition **4**
3.1 Dates and match quality 4
3.2 Author metadata and linkage audit 5
3.3 Deterministic broad genre labels 7

4 Publication-year validation **8**
4.1 Validation sources 10
4.2 Error taxonomy 10

5 Reuse demonstration: signal in book text **11**
5.1 Complete model results 12
5.1.1 Genre 12
5.1.2 Period 12
5.1.3 Year 12
5.2 Confusion matrices 13
5.2.1 Genre 13
5.2.2 Period 13
5.2.3 Year 14

6 Supplementary data files **15**

7 Limitations **16**

8 Computational provenance **16**

1. Scope and reproducibility

This supplement documents the frozen Gutenberg Atlas release and provides a worked demonstration of its use as a text corpus. The canonical release has one row per Project Gutenberg book and is generated uniformly by the build pipeline. Manual validation evaluates that pipeline; it does not patch individual production rows.

All descriptive tables below are calculated from the frozen release files. The reuse demonstration is rendered from held-out predictions and bootstrap results generated by `02-models.Rmd`. Re-running that source report downloads the sampled public-domain texts, reconstructs the text views, fits the models, and overwrites the generated result files. Rendering this supplement itself does not download texts or refit models.

The counts characterize books selected from Project Gutenberg. They are not estimates of the historical population of English-language books or authors.

2. Corpus construction

An eligible record is an English-language Project Gutenberg text with one named personal author. Publication year is estimated from the earliest HathiTrust record matched on normalized author surname and title. A record is released only if its Project Gutenberg agent identifier resolves to a Wikidata entity explicitly recorded as human and the matched year passes the lifespan screen: at least 15 years after birth and no more than five years after death. When death is unavailable, birth plus 90 years supplies the upper bound. The release excludes matched years before 1700; later years are retained and their lower validation accuracy is reported.

Table 1. Corpus-construction funnel.

Decision	Books
release	21,296
no match	9,535
no wikidata link	5,208
review (posthumous)	2,251
review (too early)	129
review (no lifespan)	84
out of scope (pre-1700)	47

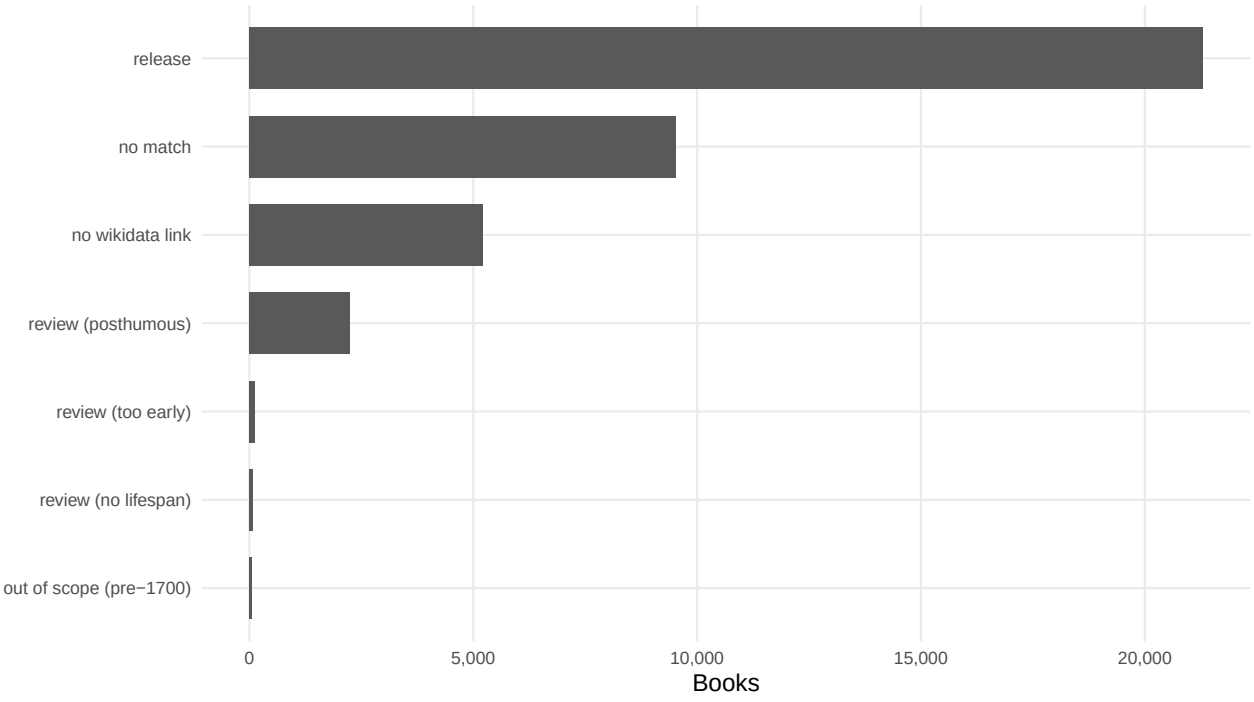


Figure 1. Corpus-construction funnel.

The pipeline releases 21,296 of 38,550 eligible records. The two largest exclusions are an unmatched HathiTrust record and the absence of a deterministic Wikidata-human link.

Exclusion is uneven by author era. Release rates vary much less by broad genre or by Wikidata gender.

Table 2. Release rates among eligible books. Gender rows include only books whose author has a Wikidata gender claim.

Variable	Group	Eligible	Released	Release rate
Author birth	Before 1750	1,057	181	17.1%
Author birth	1750-1799	1,878	1,012	53.9%
Author birth	1800-1849	12,114	7,922	65.4%
Author birth	1850-1899	16,862	11,330	67.2%
Author birth	1900 or later	1,409	178	12.6%
Author birth	Unknown	5,230	673	12.9%
Broad genre	Fiction	17,113	9,721	56.8%
Broad genre	Nonfiction	16,321	8,565	52.5%
Broad genre	Poetry	1,699	1,071	63.0%
Broad genre	Drama	826	416	50.4%
Wikidata gender	Male	24,159	16,701	69.1%
Wikidata gender	Female	6,268	4,490	71.6%

3. Release composition

3.1 Dates and match quality

`ht_year` is the earliest year among accepted HathiTrust candidates. It is an estimate of first publication, not a claim that the selected holding is a first edition. Matched years run from 1700 to 2004.

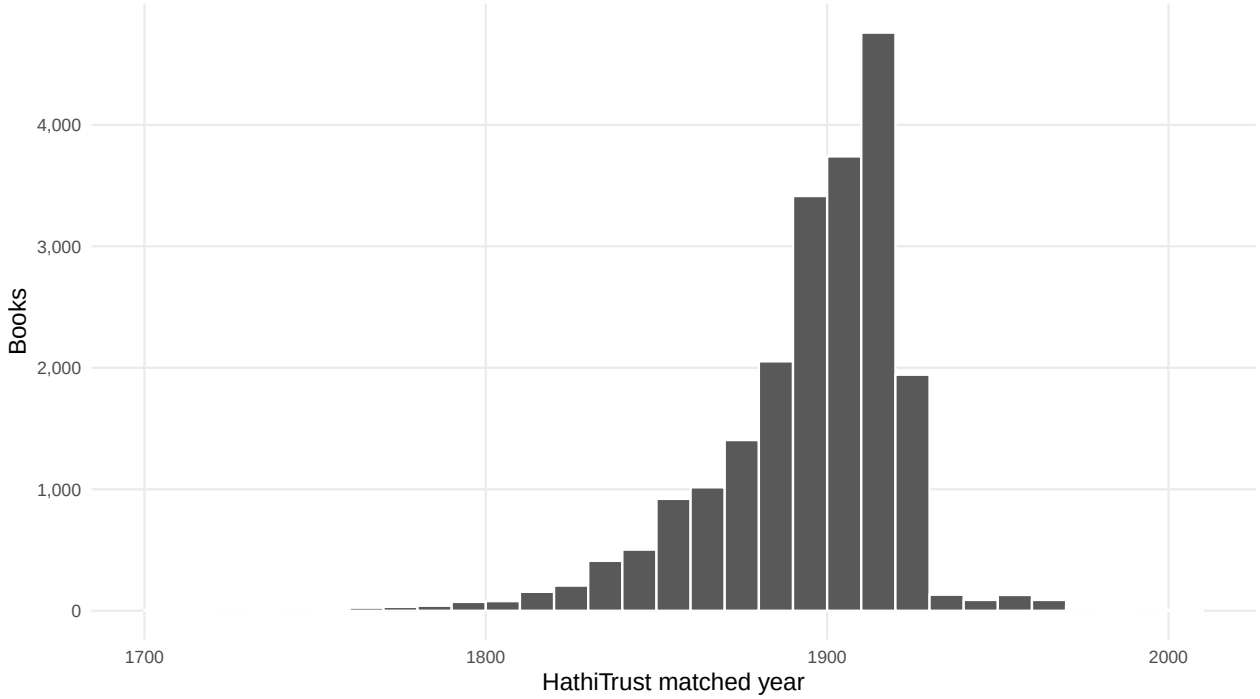


Figure 2. Distribution of matched publication years.

Table 3. HathiTrust match tiers.

Match tier	Books	Definition
1	10,674	Exact normalized title within author surname
2	10,622	Normalized-title prefix containment within author surname

3.2 Author metadata and linkage audit

Gender and country of citizenship are raw Wikidata P21 and P27 labels from the dated snapshot. They are structured claims, not inferred attributes. Coverage is measurable, but the values have not been independently validated against a second biographical authority. Project Gutenberg and Wikidata lifespan fields are retained separately; disagreements remain visible rather than being silently resolved.

Raw citizenship labels name historical states, such as the Kingdom of Great Britain or the Russian Empire, and often list the same country under several names. `citizenship_modern` maps each of the 174 recorded labels to a present-day state using `build/country_map.csv`, which assigns each historical state to the successor that holds its core territory. Twenty-seven labels involve a judgment call and are flagged in that file. The largest is the Russian Empire (83 released books), which includes authors from present-day Poland, Lithuania, Ukraine, and Finland.

Table 4. Most frequent present-day states. Books with several states count once for each.

Present-day state	Books
United States	9,066
United Kingdom	8,969
Canada	477
France	350
Ireland	311
Australia	146
Germany	133
Russia	90
India	56
New Zealand	45

Table 5. Wikidata field coverage.

Unit	Field	Available	Total	Coverage
Book	gender	21,194	21,296	99.5%
Book	citizenship	18,978	21,296	89.1%
Book	Wikidata birth-year	21,223	21,296	99.7%
Book	Wikidata death-year	21,153	21,296	99.3%
Author	gender	7,293	7,383	98.8%
Author	citizenship	5,906	7,383	80%
Author	English label	7,138	7,383	96.7%

Table 6. Cross-source linkage and lifespan diagnostics.

Diagnostic	Agree	Compared	Percent	Note
normalized name agreement	4,079	7,383	55.2%	Strict full-name token equality; diagnostic only
surname token agreement	6,981	7,383	94.6%	PG surname tokens present in Wikidata label; diagnostic only
birth year exact agreement	6,625	6,955	95.3%	Agreement, not independent accuracy
death year exact agreement	6,696	6,774	98.8%	Agreement, not independent accuracy

The release contains 7,383 resolved Project Gutenberg agents. Surname-token agreement is a linkage diagnostic, and lifespan agreement is cross-source corroboration; neither is an independent accuracy estimate for Wikidata.

Table 7. Author concentration.

Statistic	Value
Median books per author	1
Mean books per author	2.9
Authors represented by one book	4,518
Books from the 100 most frequent authors	4,253

Table 8. Ten most frequent authors in the release. Age at publication is matched publication year minus birth year; the middle range spans the 25th to 75th percentiles.

Author	Country	Books	Median age	Middle range
Edward Bulwer-Lytton	United Kingdom	203	40	34–60
Margaret Oliphant	United Kingdom	122	54	45–62
George Meredith	United Kingdom	104	60	58–63
Mark Twain	United States	101	57	46–64
Sir Gilbert Parker, 1st Baronet	United Kingdom	101	39	34–48
John Lothrop Motley	United States	99	46	42–46
William Dean Howells	United States	75	56	49–63
Horatio Alger	United States	70	49	40–61
H. G. Wells	United Kingdom	68	47	38–54
Robert Michael Ballantyne	United Kingdom	67	52	41–59

Table 9. Gender claims at book level.

Raw Wikidata gender claim	Books	Percent
male	16,701	78.4%
female	4,490	21.1%
missing	102	0.5%
cisgender woman	3	0%

Table 10. Country-of-citizenship claims at book level.

Citizenship group	Books	Percent
United States	8,571	40.2%
United Kingdom of Great Britain and Ireland	3,868	18.2%
Multiple claims	3,832	18%
Missing	2,318	10.9%
United Kingdom	1,438	6.8%
Other single claim	428	2%
Canada	407	1.9%
France	178	0.8%
Ireland	100	0.5%
Australia	86	0.4%
Kingdom of Great Britain	70	0.3%

Historical country labels and multiple P27 claims are preserved. Thus, for example, the United Kingdom and the United Kingdom of Great Britain and Ireland remain distinct rather than being collapsed into a modern nationality.

3.3 Deterministic broad genre labels

The canonical release does not add a single genre field. For description and the reuse demonstration, Project Gutenberg bookshelves, subjects, and Library of Congress classes supply four mutually exclusive analysis

labels. Fiction, poetry, and drama require one unambiguous creative-form signal. Nonfiction requires no creative-form signal and a primary LoC class outside literature. Conflicting or incomplete cases remain `excluded`.

Table 11. Deterministic broad genre labels.

Analysis label	Books	Percent
fiction	9,721	45.6%
nonfiction	8,623	40.5%
excluded	1,465	6.9%
poetry	1,071	5%
drama	416	2%

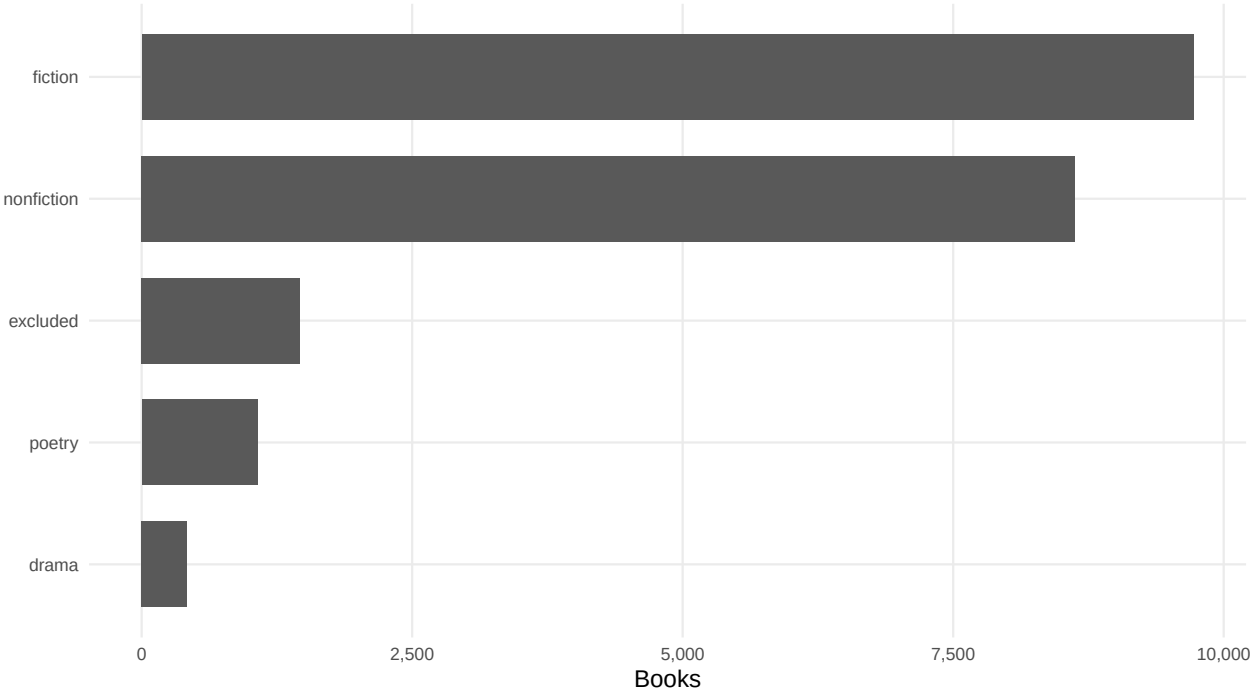


Figure 3. Distribution of deterministic broad genre labels.

4. Publication-year validation

The validation sample contains 200 release books, stratified by matched period and match tier, with at least eight sampled books per cell and the remaining positions allocated in proportion to cell size (largest-remainder rounding). Within each cell, books were drawn at random without replacement using seed 20261021 (`build/15-validation-sample.R`). Author gender was not a stratifying variable. Each resolved row has a cited reference year. A match is accurate when the HathiTrust estimate is within one year of that cited year. One composite Kipling edition remains ambiguous and receives no verdict.

The 16 sampled books dated 1925 or later had lower accuracy than the earlier periods, most often because the earliest HathiTrust holding was a later reprint. They remain part of the release and the weighted estimate.

The overall estimate combines within-stratum sample proportions using each period-by-tier cell's share of the release. The early and late periods were deliberately oversampled, so the weighted estimate—not the raw proportion—is the headline result.

Table 12. Stratified validation allocation and represented release counts.

Period	Match tier	Sampled books	Judged books	Books represented
1700-99	1	8	8	66
1700-99	2	8	8	140
1800-49	1	8	8	604
1800-49	2	8	8	717
1850-99	1	37	36	4,537
1850-99	2	31	31	3,877
1900-24	1	40	40	4,895
1900-24	2	44	44	5,522
1925+	1	8	8	572
1925+	2	8	8	366

Table 13. Weighted publication-year accuracy.

Judged	Accuracy within one year	95% interval
199	87.4%	82.7% to 92.1%

Table 14. Accuracy by validation period.

Period	Judged	Accuracy
1700-99	16	81.2%
1800-49	16	81.2%
1850-99	67	89.6%
1900-24	84	89.3%
1925+	16	62.5%

Table 15. Absolute error by validation period.

period	Books	Within one year	Within five years	Median absolute gap
1700-99	16	81.2%	93.8%	0
1800-49	16	81.2%	87.5%	0
1850-99	67	89.6%	94%	0
1900-24	84	89.3%	92.9%	0
1925+	16	62.5%	75%	0

4.1 Validation sources

Sources for the validation years were chosen book by book, without a fixed source hierarchy. Each cited source was classified after the fact by type (`build/20-validation-evidence.R`). Catalogue records and digitized editions date a particular edition. They establish first publication only when the source note records first-edition evidence. Reference works, contemporary reviews, general web pages, and trade listings assert first publication directly. HathiTrust catalogue records, the same catalogue used for matching, are cited for 23 dates.

Table 16. Validation years by evidence class.

Evidence class	Judged	Asserts first publication	Accuracy (unweighted)
Catalogue record	55	2	83.6%
General web	52	52	82.7%
Digitized edition	40	0	92.5%
Trade listing	27	27	88.9%
Reference or bibliography	19	19	84.2%
Contemporary review or serial	6	6	83.3%

Table 17. Weighted accuracy under source restrictions. Estimates with fewer than eight cells are reweighted over the represented cells.

Subset	Judged	Cells	Weighted accuracy	95% interval
All confirmed	199	10	87.4%	82.7% to 92.1%
Excluding HathiTrust catalogue	176	10	87.4%	82.3% to 92.5%
Excluding trade listings	172	10	86.3%	81.2% to 91.5%
First-publication claims only	106	10	86.2%	78.8% to 93.5%
Excluding trade and general web	120	9	86.4%	80.2% to 92.7%

No source restriction moves the weighted estimate by more than 1.2 points. The reported accuracy therefore does not depend on the weaker source types.

4.2 Error taxonomy

The 28 errors consist of later-edition or reprint dates that pass the lifespan screen, multi-volume records for which the HathiTrust year refers to the set rather than the component represented by Project Gutenberg, and one wrong-book match. The categories diagnose failure modes; they are not used to manually correct the release.

Table 18. Taxonomy of publication-year errors.

Failure mode	Books
Later edition or reprint	24
Multi-volume component/set mismatch	3
Wrong-book match	1

Later editions and reprints are the dominant failure mode, including most errors among matched years after 1924.

These errors have two consequences for analyses based on author lifespans. First, they are asymmetric. In the validation sample, 25 of the 28 errors place a book later than its first publication, so ages at publication skew older and an author's latest dates are the least reliable. Second, the lifespan screen truncates ages by construction: no released book is dated before its author's fifteenth birthday or more than five years after death. Age distributions near either bound therefore reflect the release rule, not publishing history. Author-level summaries should use medians or percentiles rather than minimum and maximum ages, as in the author table above. A future improvement should be a general, pre-specified rule evaluated on a new targeted sample, not a ledger of manual corrections.

5. Reuse demonstration: signal in book text

This demonstration asks whether a passage from each book carries information recorded in the metadata. It is evidence that the release can support corpus research, not an additional validation of individual metadata values.

Three held-out tasks predict broad genre, 25-year period, and publication year. The date tasks use 1800–1924 as a fixed analytical window; this is not a release boundary. Five-fold evaluation is author-disjoint, so all books by an author remain in the same fold.

Table 19. Reuse-demonstration samples.

Task	Class	Books	Authors
genre	drama	120	96
genre	fiction	120	82
genre	nonfiction	120	97
genre	poetry	120	92
period	1850-1874	120	91
period	1875-1899	120	88
period	1900-1924	120	90
year	1800-1824	80	60
year	1825-1849	80	62
year	1850-1874	80	62
year	1875-1899	80	62
year	1900-1924	80	56

Each book contributes up to 5,000 words from its body after the Gutenberg header and the first 400 post-marker words. Word and character TF–IDF models use penalized regression; averaged BGE embeddings

use ridge regression. Raw opening-text models act as a front-matter check. Classification is evaluated with macro-F1 and balanced accuracy; numeric year prediction uses mean absolute error and correlation. Intervals come from 1,000 author-cluster bootstrap samples.

Table 20. Best body-text result for each task.

Task	Best body model	Baseline	Body score
Genre	Body, BGE	0.189 macro-F1	0.865 macro-F1
Period	Body, BGE	0.308 macro-F1	0.441 macro-F1
Year	Body, words	30.9 years MAE	20.2 years MAE

Genre is strongly recoverable from body text. Period and continuous year also show signal, though raw front matter improves temporal prediction.

5.1 Complete model results

5.1.1 Genre

Table 21. Complete held-out results: genre

Model	Macro-F1 [95% CI]	Gain	Balanced accuracy
Baseline	0.189 [0.153, 0.222]		0.225
Raw opening, words	0.585 [0.532, 0.636]	0.396 [0.347, 0.450]	0.571
Raw opening, characters	0.805 [0.764, 0.840]	0.616 [0.561, 0.667]	0.804
Body, words	0.818 [0.779, 0.852]	0.629 [0.581, 0.678]	0.817
Body, characters	0.803 [0.761, 0.837]	0.614 [0.562, 0.662]	0.802
Body, BGE	0.865 [0.831, 0.898]	0.676 [0.629, 0.723]	0.865

5.1.2 Period

Table 22. Complete held-out results: period

Model	Macro-F1 [95% CI]	Gain	Balanced accuracy
Baseline	0.308 [0.252, 0.366]		0.314
Raw opening, words	0.529 [0.476, 0.578]	0.221 [0.142, 0.303]	0.525
Raw opening, characters	0.342 [0.291, 0.398]	0.034 [-0.031, 0.103]	0.347
Body, words	0.365 [0.313, 0.417]	0.057 [-0.011, 0.121]	0.364
Body, characters	0.411 [0.357, 0.461]	0.103 [0.034, 0.168]	0.417
Body, BGE	0.441 [0.388, 0.495]	0.133 [0.056, 0.214]	0.442

5.1.3 Year

Table 23. Complete held-out results: year

Model	Mean absolute error [95% CI]	Gain	Correlation
Baseline	30.9 [28.8, 33.0]		-0.02
Raw opening, words	16.7 [15.5, 18.1]	14.1 [12.4, 16.0]	0.82
Raw opening, characters	19.2 [17.7, 20.6]	11.7 [9.8, 13.8]	0.74
Body, words	20.2 [18.6, 21.7]	10.7 [9.0, 12.5]	0.73
Body, characters	21.8 [20.2, 23.4]	9.0 [7.2, 11.0]	0.68
Body, BGE	21.6 [20.1, 23.3]	9.3 [7.7, 10.8]	0.70

5.2 Confusion matrices

5.2.1 Genre

5.2.1.1 Body, words

Actual	drama	fiction	nonfiction	poetry	Recall
drama	92	3	16	9	0.77
fiction	4	103	9	4	0.86
nonfiction	2	20	98	0	0.82
poetry	6	4	11	99	0.82

5.2.1.2 Body, characters

Actual	drama	fiction	nonfiction	poetry	Recall
drama	94	3	16	7	0.78
fiction	4	101	11	4	0.84
nonfiction	6	20	92	2	0.77
poetry	10	5	7	98	0.82

5.2.1.3 Body, BGE

Actual	drama	fiction	nonfiction	poetry	Recall
drama	97	3	11	9	0.81
fiction	2	108	9	1	0.90
nonfiction	5	13	102	0	0.85
poetry	6	3	3	108	0.90

5.2.2 Period

5.2.2.1 Body, words

Actual	1850-1874	1875-1899	1900-1924	Recall
1850-1874	50	39	31	0.42
1875-1899	42	38	40	0.32
1900-1924	23	54	43	0.36

5.2.2.2 Body, characters

Actual	1850-1874	1875-1899	1900-1924	Recall
1850-1874	60	24	36	0.50
1875-1899	41	33	46	0.28
1900-1924	31	32	57	0.48

5.2.2.3 Body, BGE

Actual	1850-1874	1875-1899	1900-1924	Recall
1850-1874	51	40	29	0.42
1875-1899	39	46	35	0.38
1900-1924	25	33	62	0.52

5.2.3 Year

5.2.3.1 Body, words

Actual	1800-1824	1825-1849	1850-1874	1875-1899	1900-1924	Recall
1800-1824	6	51	23	0	0	0.07
1825-1849	7	27	42	4	0	0.34
1850-1874	1	6	62	8	3	0.78
1875-1899	1	3	35	33	8	0.41
1900-1924	0	0	18	46	16	0.20

5.2.3.2 Body, characters

Actual	1800-1824	1825-1849	1850-1874	1875-1899	1900-1924	Recall
1800-1824	8	47	23	2	0	0.10
1825-1849	3	31	37	9	0	0.39
1850-1874	2	10	44	22	2	0.55
1875-1899	0	3	39	37	1	0.46
1900-1924	0	1	19	47	13	0.16

5.2.3.3 Body, BGE

Actual	1800-1824	1825-1849	1850-1874	1875-1899	1900-1924	Recall
1800-1824	0	54	25	1	0	0.00
1825-1849	0	37	43	0	0	0.46
1850-1874	0	15	43	22	0	0.54
1875-1899	0	5	41	32	2	0.40
1900-1924	0	0	21	48	11	0.14

6. Supplementary data files

The following machine-readable files accompany this report:

Table 33. Supplementary file manifest.

File	Role
build/data/release.csv	Canonical 21,296-book release
build/data/unverified_candidates.csv	Excluded records and decision reasons
build/data/funnel.csv	Corpus-construction counts
build/data/author_metadata_summary.csv	Author-field coverage and linkage diagnostics
build/data/date_validation_summary.csv	Weighted and period-specific date validation
build/data/validation_sourced.csv	Cited first-publication truth for the validation sample
build/data/validation_evidence.csv	Evidence class and first-publication claim for each validation source
analysis/validation_audit.csv	Human-readable joined validation audit
analysis/results/model_sample.csv	Author-disjoint modeling sample and folds
analysis/results/model_metrics.csv	Held-out secondary model metrics
analysis/results/model_bootstrap.csv	Author-cluster bootstrap estimates
analysis/results/predictions_[task].csv	Held-out predictions for genre, period, and year

7. Limitations

The archive is shaped by public-domain law, volunteer selection, cataloguing, and survival; it is not a representative sample of historical publishing. HathiTrust dates sometimes identify a reprint or set rather than first book publication. Wikidata demographic fields are dated public claims with uneven coverage, especially for citizenship, and they should not be treated as independently verified personal attributes. Broad genre labels are analysis conveniences derived deterministically from catalog metadata. Finally, the reuse models show association in this selected corpus and do not identify essential properties of genres, periods, or author identities.

8. Computational provenance

```
## R version 4.3.1 (2023-06-16)
## Platform: x86_64-conda-linux-gnu (64-bit)
## Running under: Ubuntu 26.04.1 LTS
##
## Matrix products: default
## BLAS/LAPACK: /home/jtimm/miniconda3/lib/libopenblasp-r0.3.30.so;  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
##  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
##  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
## [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
```

```
##
## time zone: America/Denver
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] knitr_1.51          ggplot2_3.5.2       data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       dplyr_1.1.4        compiler_4.3.1     tidyselect_1.2.1
##  [5] tinytex_0.57       xml2_1.5.2         stringr_1.6.0      systemfonts_1.3.2
##  [9] scales_1.4.0       textshaping_1.0.5  yaml_2.3.7         fastmap_1.2.0
## [13] R6_2.6.1           labeling_0.4.3     generics_0.1.4     tibble_3.2.1
## [17] kableExtra_1.4.1   svglite_2.2.2      pillar_1.11.0      RColorBrewer_1.1-3
## [21] rlang_1.1.7        stringi_1.8.7      xfun_0.60          otel_0.2.0
## [25] viridisLite_0.4.2  cli_3.6.5          withr_3.0.2        magrittr_2.0.3
## [29] digest_0.6.33      grid_4.3.1         rstudioapi_0.18.0  lifecycle_1.0.4
## [33] vctrs_0.6.4        evaluate_1.0.5     glue_1.8.0         farver_2.1.1
## [37] rmarkdown_2.31     tools_4.3.1        pkgconfig_2.0.3    htmltools_0.5.9
```